Section B - Speed Math: Questions & Answers

Futures First Test Paper

Question 1: 88888 + 8888 + 888 + 88 + 8 =

Answer: 98760

Working: 88888+8888=97776; +888=98664; +88=98752; +8=98760

Question 2: 105 + 83 + 122 =

Answer: 310

Working: 105+83=188; 188+122=310

Question 3: $28^3 + 17^3 = 10 + x^3$. Find x (round to 2 decimal places)

Answer: $x \approx 29.95$

Working: 28^3=21952, 17^3=4913. Sum = 26865. 26865-10=26855. x = cube root of 26855 ≈ 29.9499 → 29.95

Question 4: 23.75 × 18.60 =

Answer: 441.75

Working: 23.75×18 = 427.5; 23.75×0.6 = 14.25; total = 441.75

Question 5: 875 × 42.80% =

Answer: 374.5

Working: 875 × 0.428 = 374.5

Question 6: 2222 × 13 / 111 =

Answer: ≈ 260.23

Working: 2222×13 = 28886; 28886 ÷ 111 = 260.234... (repeating), rounded ≈ 260.23

Question 7: In a cricket match, a team scored 282 runs in 18 overs and 3 balls. Calculate the run rate (round to 2 decimal places).

Answer: 15.24

Working: 18 overs 3 balls = 18.5 overs (3 balls = 0.5 over). Run rate = 282 ÷ 18.5 = 15.2432... ≈ 15.24

Question 8: Find the square root of 12321 =

Answer: 111

Working: 111 × 111 = 12321

Question 9: $(45.6)^2 - (44.4)^2$ =

Answer: 108

Working: Using $a^2-b^2 = (a-b)(a+b)$: (45.6–44.4)(45.6+44.4) = 1.2 × 90 = 108

Question 10: 8750 ÷ 3.5 + 6250 ÷ 2.5 =

Answer: 5000

Working: 8750÷3.5 = 2500; 6250÷2.5 = 2500; total = 5000

Question 11: A sum of ■75,000 is invested at 8.75% p.a. simple interest for 2.5 years. Find the interest earned.

Answer: ■16,406.25

Working: SI = P×R×T/100 = 75000 × 8.75 × 2.5 / 100 = 75000 × 21.875 / 100 = 16,406.25

Question 12: $12^3 + 18^3 - 24^3$ =

Answer: –6264

Working: 12^3=1728, 18^3=5832, 24^3=13824. 1728+5832=7560; 7560–13824 = –6264

Question 13: $999^2 - 998^2$ =

Answer: 1997

Working: Using $a^2-b^2 = (a-b)(a+b)$: (999–998)(999+998) = 1 × 1997 = 1997

Question 14: A trader gives 12.5% discount on a marked price of ■4800 and still gains 20%. Find the cost price.

Answer: ■3,500

Working: Selling Price = 4800 × (1–0.125) = 4800 × 0.875 = 4200. Cost Price = SP ÷ 1.20 = 4200 ÷ 1.2 = 3500

Question 15: If 1 USD = ■83.75, convert $2450 into INR.

Answer: ■2,05,187.50

Working: 2450 × 83.75 = 2450×80 + 2450×3.75 = 196000 + 9187.5 = 205187.5